

5.1.77

- a. Prove that $n! + 2$ is divisible by 2, for every integer $n \geq 2$.
- b. Prove that $n! + k$ is divisible by k , for every integer $n \geq 2$ and $k = 2, 3, \dots, n$.

Proof:

Suppose $n \geq 2, \forall n \in \mathbb{Z}$. We must show that $n! + 2$ is divisible by 2.

$$\text{Let } P(n) = \frac{n! + 2}{2}$$

Basic step: $P(2)$

$$P(2) = \frac{2! + 2}{2} = \frac{4}{2} = \boxed{2}$$

Therefore, $P(2)$ is true.

Inductive step: $P(k)$

Suppose $P(k)$ is true for any $k \in \mathbb{Z}, k \geq 2$, then:

$$P(k) = \frac{k! + 2}{2} \in \mathbb{Z}$$

We must show that $P(k + 1)$ is true.

$$\begin{aligned} P(k + 1) &= \frac{(k + 1)! + 2}{2} = \\ &= \frac{(k + 1)k! + 2}{2} \end{aligned}$$

$$\frac{k! + 2}{2} = m, \quad m \in \mathbb{Z}$$

$$k! + 2 = 2m$$

$$k! = 2m - 2$$

$$\begin{aligned} P(k + 1) &= \frac{(k + 1)(2m - 2) + 2}{2} = \\ &= \frac{2km - 2k + 2m - 2 + 2}{2} = \\ &= \frac{2km - 2k + 2m}{2} = \boxed{km - k + m \in \mathbb{Z}} \end{aligned}$$

Therefore, $P(k + 1)$ is true, which is what was to be shown.

□

Solution: a. $n! + 2$ is divisible by 2, for every integer $n \geq 2$.

b. Prove that $n! + k$ is divisible by k , for every integer $n \geq 2$ and $k = 2, 3, \dots, n$.

Proof:

Suppose $n \geq 2, \forall n \in \mathbb{Z}$. We must show that $n! + k$ is divisible by $k \in \mathbb{Z}$, i.e. $k = 2, 3, \dots, n$.

$$\text{Let } P(n) = \frac{n! + k}{k}$$

Since $k \leq n$, k is one of the factors inside $n!$. For example if $n = 6$ and $k = 4$, then:

$$6! = 6 \times 5 \times \boxed{4} \times 3!$$

Hence, $4 \mid 6!$. In general:

$$n! = n(n-1)\dots k \dots 2 \times 1$$

Thus, $k \mid n!$. That means $\exists m \in \mathbb{Z}$ such that $n! = km$.

$$\begin{aligned} \text{Let } n! &= km \\ n! + k &= km + k \\ &= k(m + 1) \end{aligned}$$

Since $m + 1 \in \mathbb{Z}$, then $k \mid (n! + k)$, which is what was to be shown.

□

Solution: b. $n! + k$ is divisible by k , for every integer $n \geq 2$.